Routing In Angular

Routing helps you change what the user sees in a single-page app. Routing, in the context of web development, refers to the process of determining how an application responds to a specific URL or path. It involves mapping URLs to different components or views within the application and rendering the appropriate content based on the requested URL.

In a client-side web application, such as a single-page application (SPA) built with Angular, routing allows users to navigate between different views without actually loading a new HTML page from the server. Instead, the application dynamically updates the content in the browser by loading the necessary components and data based on the requested route.

In a single page application, click on hyperlinks doesn't reload whole DOM, instead URL and component is loaded. This will prevent static resources like CSS, js , html to reload again and again thus improving page loading time and improve user experience.

Routing in Angular typically involves the following features:

Routes: Routes define the mapping between the URL path and the corresponding components to be rendered. Each route is defined with a URL path and a corresponding component to be displayed when that path is accessed.

Router: The router is responsible for interpreting the current URL and loading the appropriate components based on the defined routes. It listens to URL changes and handles navigation within the application.

Router Outlet: The router outlet is a placeholder in the application's template where the content of the current route is rendered.

Router Links and Navigation: Links and navigation elements, such as anchor tags (<a>) or buttons, are used to trigger navigation to different routes within the application. These elements can be decorated with directives like routerLink in Angular to specify the target route.

The following command uses the Angular CLI to generate a basic Angular application with application routes.

>ng new routingDemo

Add components for routing

To use the Angular router, an application needs to have at least few components so that it can navigate from one to the other. To create a component using the CLI, enter the following at the command line where first is the name of your component:

>ng generate component first

>ng generate component second

Angular by Ranjan B.

Here we are working with a CLI-generated Angular application. Angular CLI will use <base href="/"> in the <head> of index.html file. This assumes that the app folder is the application root, and uses "/".

Defining a basic route

There are three fundamental building blocks to creating a route. Import the routes into app.config.ts and add it to the provideRouter function. The following is the default ApplicationConfig using the CLI.

```
export const appConfig: ApplicationConfig = {
  providers: [provideZoneChangeDetection({ eventCoalescing: true }),
provideRouter(routes)]
};
```

The Angular CLI performs this step for you. However, if you are creating an application manually or working with an existing, non-CLI application, verify that the imports and configuration are correct.

Now follow 3 steps:

1. Set up a Routes array for your routes
2. Define your routes in your Routes array
3. Add your routes to your application

1. Routes array is automatically created while creating application using Angular CLI. In the app component it has created app.routes.ts file and it has:

```
import { Routes } from '@angular/router';

export const routes: Routes = [];
```

2. Each route in this array is a JavaScript object that contains two properties. The first property, path, defines the URL path for the route. The second property, component, defines the component Angular should use for the corresponding path. Modify app.routes.ts as:

```
import { Routes } from '@angular/router';
import { FirstComponent } from './first/first.component';
import { SecondComponent } from './second/second.component';

export const routes: Routes = [
    { path: 'first-component', component: FirstComponent },
    { path: 'second-component', component: SecondComponent },
];
```

3. Now here we have defined our routes, add them to application. First, add links to the two components. Assign the anchor tag that you want to add the route to the **routerLink** attribute. Set the value of the attribute to the component to show when a user clicks on each link. Next, update your component template to include <router-outlet>. This element informs Angular to update the application view with the component for the selected route.

Update app.component.html

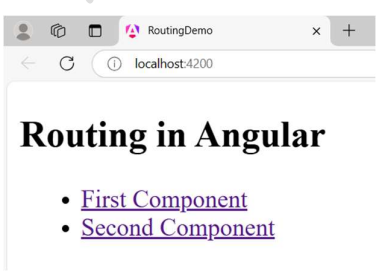
```
<h2>Routing in Angular</h2>
<nav>
  <ul>
    <li>
      <a
        routerLink="/first-component"
        routerLinkActive="active"
        ariaCurrentWhenActive="page"
        >First Component</a
      >
    </li>
    <li>
      <a
        routerLink="/second-component"
        routerLinkActive="active"
        ariaCurrentWhenActive="page"
        >Second Component</a
      >
    </li>
  </ul>
</nav>
<router-outlet />
```

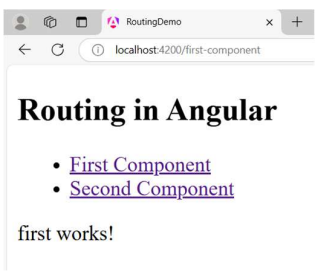
You also need to add the RouterLink, RouterLinkActive, and RouterOutlet to the imports array of AppComponent. Go to app.component.ts and add:

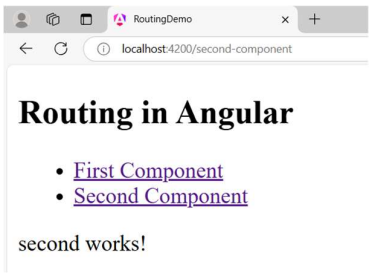
```
  imports: [RouterLink, RouterOutlet, RouterLinkActive],
```

Route order

The order of routes is important because the Router uses a first-match wins strategy when matching routes, so more specific routes should be placed above fewer specific routes. List routes with a static path first, followed by an empty path route, which matches the default route. The wildcard route comes last because it matches every URL and the Router selects it only if no other routes match first.







Setting up default route / first route

To set up a redirect, configure a route with the path you want to redirect from, the component you want to redirect to, and a pathMatch value that tells the router how to match the URL.

Here following route defined as first route in Routes array in app.routes.ts, so that as application starts by default first-component should load.

```
{ path: '', redirectTo: '/first-component', pathMatch: 'full' },
```

In the above example, the route is a redirect so that the router defaults to the first-component route. Notice that this redirect precedes the wildcard route which is used at last. Here, path: '' means to use the initial relative URL ('').

Go to app.component.html and add another link which is broken link, means it doesn't contain any component and we want to test wildcard '**' route. So, if anyone use this link it is broken request so going to wildcard request and there, we want to show 404 Not Found.

```
<li>
      <a
        routerLink="/third-component"
        routerLinkActive="active"
        ariaCurrentWhenActive="page"
        >Third Component
      </a>
</li>
```

Setting up wildcard routes

A well-functioning application should gracefully handle when users attempt to navigate to a part of your application that does not exist. To add this functionality to your application, you set up a wildcard route. The Angular router selects this route any time the requested URL doesn't match any router paths. To set up a wildcard route, add the following code to your routes definition.

{ path: '**', component: <component-name> }

The two asterisks, **, indicate to Angular that this routes definition is a wildcard route. For the component property, you can define any component in your application. Common choices include an application-specific PageNotFoundComponent, which you can define to display a 404 page to your users; or a redirect to your application's main component. A wildcard route is the last route because it matches any URL.

To display a 404 page, set up a wildcard route with the component property set to the component you'd like to use for your 404 page. Go to terminal and add PageNotFoundComponent.

>ng g c PageNotFound

Modify .html and .css with a good template as:

page-not-found.component.css

```
.cls-1 {
  fill: #ffc541;
}
.cls-2 {
  fill: #4e4066;
}
.cls-3 {
  fill: #6f5b92;
}
.cls-4 {
  fill: #f78d5e;
}
.cls-5 {
  fill: #fa976c;
}
.cls-6,
.cls-7,
.cls-8 {
  fill: #b65c32;
}
.cls-10,
.cls-6 {
  opacity: 0.6;
}
.cls-7 {
  opacity: 0.4;
}
.cls-9 {
  fill: #f4b73b;
}
.cls-11 {
  fill: #f9c358;
}
.cls-12 {
  fill: #9b462c;
}
.cls-13 {
  fill: #aa512e;
```

```
}
.cls-14 {
  fill: #7d6aa5;
}
/* animations */
.wheel {
  animation: wheel-rotate 6s ease infinite;
  transform-origin: center;
  transform-box: fill-box;
}
@keyframes wheel-rotate {
  50% {
    transform: rotate(360deg);
    animation-timing-function: cubic-bezier(0.55, 0.085, 0.68,
0.53);
  }
  100% {
    transform: rotate(960deg);
  }
}

.clock-hand-1 {
  animation: clock-rotate 3s linear infinite;
  transform-origin: bottom;
  transform-box: fill-box;
}
.clock-hand-2 {
  animation: clock-rotate 6s linear infinite;
  transform-origin: bottom;
  transform-box: fill-box;
}
@keyframes clock-rotate {
  100% {
    transform: rotate(360deg);
  }
}
#box-top {
  animation: box-top-anim 2s linear infinite;
  transform-origin: right top;
  transform-box: fill-box;
}
```

```
@keyframes box-top-anim {
  50% {
    transform: rotate(-5deg);
  }
}
#umbrella {
  animation: umbrella-anim 6s linear infinite;
  transform-origin: center;
  transform-box: fill-box;
}
@keyframes umbrella-anim {
  25% {
    transform: translateY(10px) rotate(5deg);
  }
  75% {
    transform: rotate(-5deg);
  }
}
#cup {
  animation: cup-rotate 3s cubic-bezier(0.455, 0.03, 0.515, 0.955)
infinite;
  transform-origin: top left;
  transform-box: fill-box;
}
@keyframes cup-rotate {
  50% {
    transform: rotate(-5deg);
  }
}
#pillow {
  animation: pillow-anim 3s linear infinite;
  transform-origin: center;
  transform-box: fill-box;
}
@keyframes pillow-anim {
  25% {
    transform: rotate(10deg) translateY(5px);
  }
  75% {
    transform: rotate(-10deg);
  }
```

```
}
#stripe {
  animation: stripe-anim 3s linear infinite;
  transform-origin: center;
  transform-box: fill-box;
}
@keyframes stripe-anim {
  25% {
    transform: translate(10px, 0) rotate(-10deg);
  }
  75% {
    transform: translateX(10px);
  }
}
#bike {
  animation: bike-anim 6s ease infinite;
}
@keyframes bike-anim {
  0% {
    transform: translateX(-1300px);
  }
  50% {
    transform: translateX(0);
    animation-timing-function: cubic-bezier(0.47, 0, 0.745, 0.715);
  }
  100% {
    transform: translateX(1300px);
  }
}
#rucksack {
  animation: ruck-anim 3s linear infinite;
  transform-origin: top;
  transform-box: fill-box;
}
@keyframes ruck-anim {
  50% {
    transform: rotate(5deg);
  }
}
.circle {
  animation: circle-anim ease infinite;
```

```
  transform-origin: center;
  transform-box: fill-box;
  perspective: 0px;
}
.circle.c1 {
  animation-duration: 2s;
}
.circle.c2 {
  animation-duration: 3s;
}
.circle.c3 {
  animation-duration: 1s;
}
.circle.c4 {
  animation-duration: 1s;
}
.circle.c5 {
  animation-duration: 2s;
}
.circle.c6 {
  animation-duration: 3s;
}
@keyframes circle-anim {
  50% {
    transform: scale(0.2) rotateX(360deg) rotateY(360deg);
  }
}
.four,
#ou {
  animation: four-anim cubic-bezier(0.39, 0.575, 0.565, 1) infinite;
}
.four.a {
  transform-origin: bottom left;
  animation-duration: 3s;
  transform-box: fill-box;
}
.four.b {
  transform-origin: bottom right;
  animation-duration: 3s;
  transform-box: fill-box;
}
```

```
#ou {
  animation-duration: 6s;
  transform-origin: center;
  transform-box: fill-box;
}
@keyframes four-anim {
  50% {
    transform: scale(0.98);
  }
}
```

page-not-found.component.html

```
<div class="wrapper">
  <svg xmlns="http://www.w3.org/2000/svg" viewBox="0 0 1920 1080">
    <title>404</title>
    <g id="Layer_12 yellow-back-fig" data-name="Layer 12">
      <path
        class="cls-1"
        d="M600.87,872H156a4,4,0,0,0-
3.78,4.19h0a4,4,0,0,0,3.78,4.19H600.87a4,4,0,0,0,3.78-
4.19h0A4,4,0,0,0,600.87,872Z"
      />
      <rect
        class="cls-1"
        x="680.91"
        y="871.98"
        width="513.38"
        height="8.39"
        rx="4.19"
        ry="4.19"
      />
      <path
        class="cls-1"
        d="M1480,876.17h0c0,2.32,2.37,4.19,5.3,4.19h350.61c2.93,0,5.
3-1.88,5.3-4.19h0c0-2.32-2.37-4.19-5.3-
4.19H1485.26C1482.33,872,1480,873.86,1480,876.17Z"
      />
      <rect
        class="cls-1"
        x="492.21"
        y="920.64"
```

```
        width="249.8"
        height="8.39"
        rx="4.19"
        ry="4.19"
      />
      <path
        class="cls-1"
        d="M1549.14,924.84h0a4.19,4.19,0,0,0-4.19-
4.19H1067.46a14.66,14.66,0,0,1,.35,3.21v1A4.19,4.19,0,0,0,1072,929h4
72.94A4.19,4.19,0,0,0,1549.14,924.84Z"
      />
      <path
        class="cls-1"
        d="M865.5,924.84h0a4.19,4.19,0,0,0,4.19,4.19h82.37a12.28,12.
28,0,0,1-.19-2v-2.17a4.19,4.19,0,0,0-4.19-4.19h-
78A4.19,4.19,0,0,0,865.5,924.84Z"
      />
      <rect
        class="cls-1"
        x="915.6"
        y="981.47"
        width="54.72"
        height="8.39"
        rx="4.19"
        ry="4.19"
      />
      <path
        class="cls-1"
        d="M730.33,985.67h0c0,2.32,4.23,4.19,9.44,4.19h104.3c5.22,0,
9.44-1.88,9.44-4.19h0c0-2.32-4.23-4.19-9.44-
4.19H739.78C734.56,981.47,730.33,983.35,730.33,985.67Z"
      />
      <rect
        class="cls-1"
        x="997.06"
        y="981.47"
        width="78.11"
        height="8.39"
        rx="4.19"
        ry="4.19"
      />
```

```
      <g id="round-conf">
        <path
          class="cls-1 circle c1"
          d="M536.41,155.14a17.77,17.77,0,1,0,17.77,17.77A17.77,17.7
7,0,0,0,536.41,155.14Zm0,28.68a10.9,10.9,0,1,1,10.9-
10.9A10.9,10.9,0,0,1,536.41,183.81Z"
        />
        <path
          class="cls-1 circle c2"
          d="M1345.09,82.44a17.77,17.77,0,1,0,17.77,17.77A17.77,17.7
7,0,0,0,1345.09,82.44Zm0,28.68a10.9,10.9,0,1,1,10.9-
10.9A10.9,10.9,0,0,1,1345.09,111.12Z"
        />
        <path
          class="cls-1 circle c3"
          d="M70.12,363A17.77,17.77,0,1,0,87.89,380.8,17.77,17.77,0,
0,0,70.12,363Zm0,28.68A10.9,10.9,0,1,1,81,380.8,10.9,10.9,0,0,1,70.1
2,391.7Z"
        />
        <path
          class="cls-1 circle c4"
          d="M170.47,751.82a17.77,17.77,0,1,0,17.77,17.77A17.77,17.7
7,0,0,0,170.47,751.82Zm0,28.68a10.9,10.9,0,1,1,10.9-
10.9A10.9,10.9,0,0,1,170.47,780.5Z"
        />
        <path
          class="cls-1 circle c5"
          d="M1457.34,762.73a17.77,17.77,0,1,0,17.77,17.77A17.77,17.
77,0,0,0,1457.34,762.73Zm0,28.68a10.9,10.9,0,1,1,10.9-
10.9A10.9,10.9,0,0,1,1457.34,791.4Z"
        />
        <path
          class="cls-1 circle c6"
          d="M1829.15,407.49a17.77,17.77,0,1,0,17.77,17.77A17.77,17.
77,0,0,0,1829.15,407.49Zm0,28.68a10.9,10.9,0,1,1,10.9-
10.9A10.9,10.9,0,0,1,1829.15,436.17Z"
        />
      </g>
    </g>
    <g id="fortyfour" data-name="Layer 2">
```

```
      <g class="four a">
        <rect
          class="cls-2"
          x="233.74"
          y="391.14"
          width="120.71"
          height="466.29"
          rx="57.1"
          ry="57.1"
          transform="translate(918.39 330.19) rotate(90)"
        />

        <rect
          class="cls-3"
          x="333.83"
          y="475.1"
          width="120.71"
          height="396.88"
          rx="60.36"
          ry="60.36"
        />

        <rect
          class="cls-3"
          x="197.13"
          y="122.89"
          width="120.71"
          height="604.75"
          rx="60.36"
          ry="60.36"
          transform="translate(290.49 -70.78) rotate(35)"
        />
      </g>
      <g class="four b">
        <rect
          class="cls-2"
          x="1558.84"
          y="391.91"
          width="120.71"
          height="466.29"
          rx="57.1"
```

```
          ry="57.1"
          transform="translate(2244.26 -994.14) rotate(90)"
        />

        <rect
          class="cls-3"
          x="1658.92"
          y="475.87"
          width="120.71"
          height="396.88"
          rx="60.36"
          ry="60.36"
        />

        <rect
          class="cls-3"
          x="1522.22"
          y="123.66"
          width="120.71"
          height="604.75"
          rx="60.36"
          ry="60.36"
          transform="translate(530.57 -830.68) rotate(35)"
        />
      </g>
      <path
        class="cls-3"
        id="ou"
        d="M956.54,168.2c-194.34,0-351.89,157.55-
351.89,351.89S762.19,872,956.54,872s351.89-157.55,351.89-
351.89S1150.88,168.2,956.54,168.2Zm0,584.49c-128.46,0-232.6-104.14-
232.6-232.6s104.14-232.6,232.6-
232.6,232.6,104.14,232.6,232.6S1085,752.69,956.54,752.69Z"
      />
    </g>
    <g id="umbrella" data-name="Layer 3">
      <g>
        <circle
          class="cls-4"
          cx="1187.53"
          cy="240.3"
```

```
          r="7.66"
          transform="translate(236.36 990.8) rotate(-49.71)"
        />
        <g>
          <path
            class="cls-5"
            d="M1219.56,359.67l55,100.52c32.7-48.48-6.87-142.43-
91.75-214.38-84.41-71.55-183-95.33-225.81-56l114.21,44.14Z"
          />
          <path
            class="cls-6"
            d="M1182.79,245.81c-84.41-71.55-183-95.33-225.81-
56l114.21,44.14Z"
          />
          <polygon
            class="cls-7"
            points="1182.79 245.81 1071.19 233.91 1219.56 359.67
1182.79 245.81"
          />
        </g>
        <polygon
          class="cls-8"
          points="1180.91 409.02 1274.54 460.19 1219.56 359.67
1071.19 233.91 956.98 189.76 1021.95 274.29 1180.91 409.02"
        />
        <g>
          <rect
            class="cls-4"
            x="997.45"
            y="358.35"
            width="175.58"
            height="5.1"
            transform="translate(108.21 955.38) rotate(-49.71)"
          />
          <rect
            class="cls-4"
            x="1028.09"
            y="399.36"
            width="21.46"
            height="32.27"
            rx="10.73"
```

```
            ry="10.73"
            transform="translate(515.04 -573.16) rotate(40.29)"
          />
        </g>
      </g>
    </g>
    <g id="pillow" data-name="Layer 4">
      <path
        class="cls-1"
        d="M754,627.07c7,.54,12.92-2.82,13.35-7.59s-4.95-9.24-12-
9.87a18.55,18.55,0,0,0-2.17,0l-74.9-81.64c0-.1,0-.19,0-.29,0-7.09-4-
12.83-8.8-12.81s-8.75,5.77-8.73,12.87c0,0,0,.09,0,.13l-50.21,46.07h-
.09c-7.06-.63-13.14,2.77-
13.57,7.59s4.87,9.16,11.85,9.84l76.08,82.92s0,0,0,.06c0,7.09,4,12.83
,8.8,12.81s8.65-5.66,8.71-12.65Z"
      />
      <path
        class="cls-9"
        d="M669.46,514.82c-4.77-.83-8.75,5.77-
8.73,12.87,0,0,0,.09,0,.13l-50.21,46.07h-.09c-7.06-.63-13.14,2.77-
13.57,7.59s4.87,9.16,11.85,9.84l76.08,82.92s0,0,0,.06c0,7.09,4,12.83
,8.8,12.81s8.65-5.66,8.71-
12.65C570.55,573,702.07,520.47,669.46,514.82Z"
      />
    </g>
    <g id="cup" data-name="Layer 7">
      <polygon
        class="cls-1"
        points="1173.69 748.21 1140.52 715.42 1195.79 647.35 1241.13
692.16 1173.69 748.21"
      />
      <polygon
        class="cls-8"
        points="1173.69 748.21 1140.52 715.42 1143.93 711.27 1177.81
744.75 1173.69 748.21"
      />
      <polygon
        class="cls-5"
        points="1194.68 731.46 1157.04 694.24 1183.8 661.7 1226.91
704.32 1194.68 731.46"
      />
```

```
      <g class="cls-10">
        <path
          class="cls-8"
          d="M1176.32,667.78h0a4.19,4.19,0,0,1,4.19,4.19v33.54a0,0,0
,0,1,0,0h-8.38a0,0,0,0,1,0,0V672a4.19,4.19,0,0,1,4.19-4.19Z"
          transform="translate(822.53 -628.67) rotate(44.67)"
        />
        <path
          class="cls-8"
          d="M1172.73,709.7l23.58-
23.85a4.19,4.19,0,0,1,5.92,0h0a4.19,4.19,0,0,1,0,5.92l-23.58,23.85Z"
        />
        <path
          class="cls-8"
          d="M1185.11,722.09l23.58-
23.85a4.19,4.19,0,0,1,5.92,0h0a4.19,4.19,0,0,1,0,5.92L1191.06,728Z"
        />
      </g>
      <path
        class="cls-5"
        d="M1197.85,660.5h45.69a5.7,5.7,0,0,1,5.7,5.7v8.32a0,0,0,0,1
,0,0h-57.09a0,0,0,0,1,0,0v-8.32A5.7,5.7,0,0,1,1197.85,660.5Z"
        transform="translate(829.53 -667.66) rotate(45)"
      />
      <path
        class="cls-8"
        d="M1191.49,664.74h53.94a5.25,5.25,0,0,1,5.25,5.25v4.79a0,0,
0,0,1,0,0h-64.44a0,0,0,0,1,0,0V670a5.25,5.25,0,0,1,5.25-5.25Z"
        transform="translate(822.83 -663.17) rotate(44.67)"
      />
    </g>
    <g id="clock" data-name="Layer 8">
      <circle
        class="cls-5"
        cx="847.7"
        cy="247.59"
        r="74.66"
        transform="translate(-32.91 314.05) rotate(-20.6)"
      />
      <circle
        class="cls-1"
```

```
        cx="847.7"
        cy="247.59"
        r="63.44"
        transform="translate(-32.91 314.05) rotate(-20.6)"
      />
      <rect
        class="cls-3 clock-hand-1"
        x="845"
        y="189.5"
        width="6.04"
        height="58"
        rx="3.02"
        ry="3.02"
      />
      <rect
        class="cls-3 clock-hand-2"
        x="845"
        y="209.5"
        width="6.04"
        height="38"
        rx="3.02"
        ry="3.02"
        transform="translate(1611.22 -230.4) rotate(130.4)"
      />
      <circle
        class="cls-3"
        cx="847.7"
        cy="247.59"
        transform="translate(-32.91 314.05) rotate(-20.6)"
        r="3"
      />
    </g>
    <g id="box" data-name="Layer 9">
      <g id="box-top">
        <polygon
          class="cls-8"
          points="569.71 382.28 653.74 329.39 747.13 320.1 679.2
369.85 569.71 382.28"
        ></polygon>
        <polygon
          class="cls-5"
```

```
          points="691.95 367.2 570.87 392.34 565.32 383.35 687.8
357.45 691.95 367.2"
        ></polygon>

        <polygon
          class="cls-5"
          points="661.54 337.48 570.87 392.34 562.42 378.92 652.25
322.38 658.12 321.34 661.54 337.48"
        ></polygon>
        <polygon
          class="cls-7"
          points="661.54 337.48 570.87 392.34 562.42 378.92 652.25
322.38 658.12 321.34 661.54 337.48"
        ></polygon>
        <polygon
          class="cls-5"
          points="747.13 320.1 661.54 337.48 652.25 322.38 738.4
307.1 747.13 320.1"
        ></polygon>
      </g>
      <path
        class="cls-5"
        d="M588.28,420.26s3.44,5.2,5.19,8l43.1,68.48,158.81-100-
43.1-68.48q-2.63-4.17-5.47-8Z"
      ></path>
      <path
        class="cls-7"
        d="M588.28,420.26s3.44,5.2,5.19,8l43.1,68.48,158.81-100-
43.1-68.48q-2.63-4.17-5.47-8Z"
      ></path>
      <rect
        class="cls-5"
        x="693.73"
        y="335.51"
        width="83.99"
        height="90.58"
        transform="translate(-89.78 450.43) rotate(-32.19)"
      ></rect>
    </g>

    <g id="rucksack" data-name="Layer 6">
```

```
      <g id="stripe">
        <path
          class="cls-12"
          d="M1200.32,473.91h0a13.74,13.74,0,0,0-18.41,7.44l-
55,129.86a14.82,14.82,0,0,0,7.13,19.21h0a13.74,13.74,0,0,0,18.41-
7.44l55-129.86A14.82,14.82,0,0,0,1200.32,473.91Z"
        />
        <path
          class="cls-13"
          d="M1202.18,606.34h0a14,14,0,0,0-16.18-11.8l-48.83,9c-
7.59,1.4-12.66,9-11.31,16.89h0a14,14,0,0,0,16.18,11.8l48.83-
9C1198.46,621.82,1203.53,614.26,1202.18,606.34Z"
        />
      </g>
      <path
        class="cls-8"
        d="M1300.86,603l-122.93,22.74-15.44-90.91c-5.75-33.86,15.89-
66.17,48.34-72.18l11.58-2.08c32.45-6,57.26,17.66,63,51.51Z"
      />
      <path
        class="cls-1"
        d="M1307,601.91l-112.32,20.78-15.9-93.61c-5.5-32.36,15.19-
63.25,46.2-69h0c31-5.74,60.62,15.85,66.12,48.21Z"
      />
      <path
        class="cls-8"
        d="M1296.76,603.8,1215,618.92l-4.89-28.77c-2.11-12.42,5.83-
24.27,17.73-26.47l38.67-7.15c11.9-2.2,23.26,6.08,25.37,18.5Z"
      />
      <path
        class="cls-5"
        d="M1296.76,603.8l-73.41,13.58-4.92-29c-2-11.62,5.45-
22.72,16.6-24.78l33.07-6.12c11.14-2.06,21.77,5.69,23.75,17.32Z"
      />
      <path
        class="cls-4"
        d="M1231.77,469.69l-13.42,2.48a10.25,10.25,0,0,0-
8,11.92l2.38,14a9.9,9.9,0,0,0,11.42,8.33l13.42-
2.48a10.25,10.25,0,0,0,8-11.92l-2.38-
14A9.9,9.9,0,0,0,1231.77,469.69Zm7.17,20.84a6.39,6.39,0,0,1-5,7.43l-
```

```
8.36,1.55a6.17,6.17,0,0,1-7.12-5.19l-1.48-8.73a6.39,6.39,0,0,1,5-
7.43l8.36-1.55a6.17,6.17,0,0,1,7.12,5.19Z"
      />
      <path
        class="cls-14"
        d="M1233.74,471.13l-13.42,2.48a10.25,10.25,0,0,0-
8,11.92l2.38,14a9.9,9.9,0,0,0,11.42,8.33l13.42-
2.48a10.25,10.25,0,0,0,8-11.92l-2.38-
14A9.9,9.9,0,0,0,1233.74,471.13Zm7.17,20.84a6.39,6.39,0,0,1-5,7.43l-
8.36,1.55a6.17,6.17,0,0,1-7.12-5.19L1219,487a6.39,6.39,0,0,1,5-
7.43l8.36-1.55a6.17,6.17,0,0,1,7.12,5.19Z"
      />
    </g>
    <g id="bike" data-name="Layer 5">
      <path
        class="cls-8 wheel"
        d="M1139.82,780.44a76.59,76.59,0,1,0-
57.9,91.53A76.59,76.59,0,0,0,1139.82,780.44Zm-
28.12,6.33a47.59,47.59,0,0,1,.83,15.8c-30.14-6.28-47.68-21.65-54.39-
52.52A47.73,47.73,0,0,1,1111.69,786.77Zm-70.46-
30.9c10.35,26.88,10.14,50.4-13.73,70.77a47.67,47.67,0,0,1,13.73-
70.77Zm34.35,88a47.55,47.55,0,0,1-34.94-5.62c16.8-20.36,41.71-
25.94,67.09-19.46A47.66,47.66,0,0,1,1075.58,843.85Z"
      />
      <path
        class="cls-8 wheel"
        d="M864.89,789.69a76.59,76.59,0,1,0-
66.13,85.78A76.59,76.59,0,0,0,864.89,789.69Zm-
28.59,3.7a47.59,47.59,0,0,1-.64,15.81c-29.43-9-45.47-26-49.3-
57.33A47.73,47.73,0,0,1,836.3,793.39ZM769,756.1c7.82,27.72,5.43,51.1
2-
20.22,69.2A47.67,47.67,0,0,1,769,756.1Zm26.06,90.78a47.55,47.55,0,0,
1-34.27-8.83c18.61-18.72,43.93-22,68.6-
13.16A47.66,47.66,0,0,1,795.06,846.88Z"
      />
      <g>
        <rect
          class="cls-1"
          x="871.39"
          y="693.37"
          width="12.87"
```

```
          height="53.21"
          transform="translate(-165.97 273.38) rotate(-16.19)"
        />
        <path
          class="cls-5"
          d="M813.93,679.35c-3.72-5.2,2.24-18.5,9.16-
16.13,33.43,11.46,73.85,10.45,73.85,10.45,8.84.15,14.44,10.34,7.27,1
5.48-14.36,8.79-33.13,17-
56.35,9.76C830.17,693.41,819.83,687.6,813.93,679.35Z"
        />
        <path
          class="cls-7"
          d="M813.93,679.35c-3.72-5.2,2.24-18.5,9.16-
16.13,33.43,11.46,73.85,10.45,73.85,10.45,8.84.15,14.44,10.34,7.27,1
5.48-14.36,8.79-33.13,17-
56.35,9.76C830.17,693.41,819.83,687.6,813.93,679.35Z"
        />
        <path
          class="cls-5"
          d="M817.15,680.06c-3.59-5,1.69-16.51,8.37-
14.22,32.3,11.09,71.41,7.83,71.41,7.83,8.54.14,17.45,9.94,7.43,15.88
-13.87,8.51-32,16.44-
54.44,9.44C832.84,693.67,822.85,688,817.15,680.06Z"
        />
      </g>
      <g>
        <circle
          class="cls-9"
          cx="1022.66"
          cy="599.55"
          r="11.57"
          transform="translate(-4.86 8.38) rotate(-0.47)"
        />
        <path
          class="cls-1"
          d="M1069.76,792.37l-34.89-96.74,1.93-.8-1.71-4.15-1.74.72-
13.26-36.76,1.27-.42-2.25-6.76,5.94-2-2.57-7.72-9.7,3.22c-11.55-
22.55,2-36.33,15-41.86l-5.57-8.81c-23,10.29-29.61,28.75-19.53,54l-
9.38,3.12,2.56,7.72,5.47-1.82,2.25,6.76,2.36-.78,13.62,37.76-
2.35,1,1.71,4.15,2.16-
```

```
.89,34.65,96.09a7.47,7.47,0,0,0,9.56,4.49h0A7.47,7.47,0,0,0,1069.76,
792.37Z"
        />
        <circle
          class="cls-11"
          cx="1027.9"
          cy="587.94"
          r="12.99"
          transform="translate(-4.77 8.42) rotate(-0.47)"
        />
      </g>
      <path
        class="cls-5"
        d="M1021.29,654l-17.73,6.15,1.72,5.59-31.41,82.36c-
19.35,32.53-66.3,36.72-75.56,16.68l-7.09-21.5L879,747.1l3.28,10.09-
94.65,33.95c-11.49,2.29-11.85,15.79-
2.61,17.84,0,0,39.11,3.66,103,9.5a92.75,92.75,0,0,0,40.89-
5.29c44.32-16.56,57.73-50.67,57.73-50.67l26.82-
67.26a1.37,1.37,0,0,1,2.53,0l1.42,3.33,17.75-7.62Z"
      />
      <path
        class="cls-7"
        d="M1021.29,654l-17.73,6.15,1.72,5.59-31.41,82.36c-
19.35,32.53-66.3,36.72-75.56,16.68l-7.09-21.5L879,747.1l3.28,10.09-
94.65,33.95c-11.49,2.29-11.85,15.79-
2.61,17.84,0,0,39.11,3.66,103,9.5a92.75,92.75,0,0,0,40.89-
5.29c44.32-16.56,57.73-50.67,57.73-50.67l26.82-
67.26a1.37,1.37,0,0,1,2.53,0l1.42,3.33,17.75-7.62Z"
      />
    </g>
  </svg>
</div>
```

Nested Routes

As your application grows more complex, you might want to create routes that are relative to a component other than your root component. These types of nested routes are called child routes. This means you're adding a second <router-outlet> to your app, because it is in addition to the <router-outlet> in AppComponent.

In this example, there are two additional child components, child-a, and child-b. Here, FirstComponent has its own <nav> and a second <router-outlet> in addition to the one in AppComponent.

Go to first.component.html and modify as:

```
<h2>First Component</h2>
<nav>
  <ul>
    <li><a routerLink="child-a">Child A</a></li>
    <li><a routerLink="child-b">Child B</a></li>
  </ul>
</nav>
<router-outlet />
```

Right-Click on first component folder and use Open in Integrated Terminal and add two components childA and childB.

.. routingDemo\src\app\first >ng g c childA

.. routingDemo\src\app\first >ng g c childB

As we are creating the first component as parent for childA and childB component, modify first.component.ts

```
imports: [RouterLink, RouterOutlet, ChildAComponent,
ChildBComponent],
```

make sure it is updated with

```
import { ChildAComponent } from './child-a/child-a.component';
import { ChildBComponent } from './child-b/child-b.component';
```

A child route is like any other route, in that it needs both a path and a component. The one difference is that you place child routes in a children array within the parent route. Now go to the app.routes.ts and modify first component route as:

```
{
        path: 'first-component', component: FirstComponent,
        children: [
            { path: '', redirectTo: 'child-a', pathMatch: 'full' },
            {
                path: 'child-a', // child route path
                component: ChildAComponent, // child route component
that the router renders
            },
            {
                path: 'child-b',
```

```
                component: ChildBComponent, // another child route
component that the router renders
            },
        ],
    },
```

Angular by Ranjan B.

More on Routing

On Button Click

Home Template

```
<p>home works!</p>
<h1>This is My Home Page</h1>
```

Test Template

```
<p>test works!</p>
<button (click)="gotoHome()">Go to Home Page</button>
```

Test Code

```
export class TestComponent {
  constructor(private router: Router) { }
  gotoHome() {
    this.router.navigate(['/home']);
  }
}
```

Use test in App.html

```
<hr />
<app-test></app-test>
```

In app.routes.ts

```
 { path: 'second-component', component: SecondComponent },
    { path: 'home', component: HomeComponent },
    { path: '**', component: PageNotFoundComponent }
```

Query Parameters

Add About as component

>ng g c comps\About

Add route of about in app.routes.ts

```
{ path: 'about', component: AboutComponent },
```

On Home Page Template

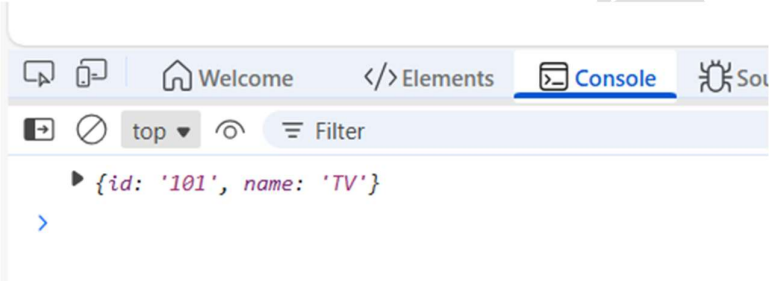
```
<a routerLink="/about">Go to About Page</a>
```

Now send some data from Home to About using Query Parameters

```
<a routerLink="/about" [queryParams]="{ id: 101, name: 'TV' }">
  Go to About Page</a>
```

Access this data on About Page

```
export class AboutComponent implements OnInit {
  constructor(public route: ActivatedRoute) { }
  ngOnInit(): void {
    this.route.queryParams.subscribe(params => {
      console.log(params);
    })
  }
}
```



Dynamic Data

Home.ts

```
export class HomeComponent {
prodData = [
  {id:1, name:'Laptop', price:45000},
  {id:2, name:'Mobile', price:25000},
  {id:3, name:'Tablet', price:15000},
  {id:4, name:'Smart Watch', price:8000},
  {id:5, name:'Wireless Earbuds', price:6000}
];
}
```

Home.html

```
Dynamic Query Params Example
<a
  routerLink="/about"
  [queryParams]="{ id: prodData[1].id, name: prodData[1].name }"
>
  Go to About Page</a
>
<br />
```

Using signals

In home.ts

```
product = signal({ id: 999, name: 'Computer', price: 70000 });
```

In home.html

```
Using Signals
<a
  routerLink="/about"
  [queryParams]="{ id: product().id, name: product().name }"
  >Get a Product</a
>
```

Get individual data on About Page

In about.ts declare respective signal to data required.

```
 pid = signal(0);
  pname = signal('');
  pprice = signal(0);
```

and use set to assign

```
this.pid.set(params['id']);
      this.pname.set(params['name']);
      this.pprice.set(params['price']);
```

In about.html

```
<h1>Product Details</h1>
<p>ID: {{ pid() }}</p>
<p>Name: {{ pname() }}</p>
<p>Price: {{ pprice() }}</p>
```

☸✠End of Chapter✠☸

Angular by Ranjan B.